

Hassan Tsuma Karungwa

Full-Stack Software Engineer · Java, Spring Boot, PostgreSQL, React, Next.js

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Available immediately · Remote (UTC+3), on-site in Kenya, or relocation

SUMMARY

Full-stack software engineer with 3+ years of independent and freelance experience designing, building and deploying complete systems: Java 21, Spring Boot and PostgreSQL on the backend, and React, Next.js and TypeScript on the frontend. Sole engineer of **RentManager**, a live multi-tenant rental platform that collects rent over M-Pesa, backed by 1,588 automated backend tests. I do my best work where correctness matters most: payments, ledgers and records a business cannot afford to get wrong.

EXPERIENCE

Full-Stack Software Engineer — Independent / Freelance Feb 2023 – Present · Remote

- Design, build and deploy full-stack systems end to end: data model, Spring Boot APIs, React / Next.js web clients, React Native mobile apps and production hosting.
- Architect multi-tenant SaaS with Domain-Driven Design and Clean Architecture, keeping business rules independent of the framework and the database.
- Build billing and payment workflows that are idempotent and transaction-safe, including M-Pesa collection and payouts.
- Test critical paths against real PostgreSQL and ship in small, focused increments with Git and CI.

PROJECTS

RentManager — Multi-tenant property management SaaS rentmanager-ke.netlify.app

Java 21 · Spring Boot 3 · PostgreSQL · Flyway · Docker · Next.js · React · TypeScript · React Native (Expo) · Clerk · M-Pesa Daraja

- Designed, built and deployed the whole product alone. Structured the backend as a modular monolith of **20 modules** (ports and adapters) covering properties, units, leases, rent, maintenance, tax, subscriptions and administration.
- Integrated **M-Pesa Daraja** in both directions, STK Push for rent collection and B2C for landlord payouts, with idempotency guards and callback handling that stays correct when callbacks arrive late, twice or out of order.
- Designed an **append-only rent ledger** in which a correction is a linked reversal, never an edit, and evolved the schema through **99 Flyway migrations**, none edited after running.
- Enforced tenant isolation from the verified JWT claim rather than a client-supplied header, at both the authorization and the repository layers.
- Wrote **1,588 backend tests** with JUnit 5, Mockito, Testcontainers and ArchUnit, covering the money paths first.
- Delivered a **63-route** Next.js web app and a **27-screen** React Native app for Android and iOS, both on a typed client generated from the backend's OpenAPI contract.
- Run it on Render, Netlify and Neon with health checks, crash reporting, GitHub Actions CI and nightly encrypted backups kept for 90 days.

whatsapp-persona-bot — AI-assisted WhatsApp auto-responder Source on GitHub

Node.js · Baileys · Anthropic Claude API · pino · pm2

- Personal tool that answers WhatsApp messages in my own writing style while I am offline, with guard rails: per-chat rate limits, randomised delays, silence once I reply, an ignore list, no prices or commitments, and a log of every message sent.

TECHNICAL SKILLS

Languages	Java, TypeScript, JavaScript, SQL
Backend	Spring Boot, Spring Security, Spring Data JPA / Hibernate, REST, OpenAPI, JWT, M-Pesa Daraja
Data	PostgreSQL, Flyway, schema and constraint design, append-only ledgers
Frontend & mobile	React, Next.js, TanStack Query, React Native (Expo) for Android and iOS
Architecture	Object-oriented design, SOLID, Clean Architecture, Domain-Driven Design, multi-tenancy, transactional outbox
Testing	JUnit 5, Mockito, Testcontainers, ArchUnit, Vitest, Jest
Delivery	Docker, Git, GitHub Actions, Render, Netlify, Neon, Agile, AI-assisted development (Claude, Codex, Aider), every change reviewed

EDUCATION

BSc Computer Science — Tom Mboya University, Homabay, Kenya Expected 2028